

# SAQIB ALI

ELECTRICAL ENGINEERING STUDENT | GROUND STATION ENGINEER

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🌐 [Portfolio](#)

## PROFILE

Electrical Engineering undergraduate at ZHCET, AMU who takes control designs from simulation to tested hardware: Ground Station Engineer at Orbitalink (a Ground Station as a Service startup) leading a two-axis satellite antenna tracker, and research intern at IIT Ropar on field-oriented control of a five-phase IPMSM. Open to roles in motor drives, control systems, embedded systems, robotics and ground station/RF engineering.

## EDUCATION

**B.Tech in Electrical Engineering, ZHCET, Aligarh Muslim University** Sep 2023 – Jun 2027

**Senior Secondary (PCM), Syedna Tahir Saifuddin (STS) School, AMU | Grade: A+** Aug 2020 – Aug 2022

## EXPERIENCE

**Research Intern | Indian Institute of Technology (IIT) Ropar** Jun 2026 – Jul 2026

- Modelled a five-phase IPMSM in MATLAB/Simulink (Clarke–Park transforms) and designed dual-plane field-oriented control with PI loops, loss-aware MTPA, flux weakening and five-phase SVPWM.
- Generated ePWM signals on a TI C2000 DSP in Code Composer Studio, tested the gate driver circuit, and submitted a research report.

**Ground Station Engineer (Part-time) | Orbitalink Private Limited, Aligarh** Aug 2025 – Present

- Lead end-to-end development and integration of a two-axis (azimuth–elevation) antenna tracking system for Orbitalink's Ground Station as a Service platform.
- Designed a circularly polarised 435 MHz cross-Yagi (MMANA-GAL/NEC, 50  $\Omega$  matching) and the tracker chassis in Fusion 360.
- Wrote firmware for NEMA23 hybrid servo motors on HSD57 drives (motion ramps, EEPROM persistence, serial command protocol); porting it from Arduino to STM32F411.

**Electrical Design Intern | Orbitalink, with AMU Innovation Foundation** Jun 2025 – Jul 2025

- Developed an adaptive PID controller on Arduino with AS5600 magnetic encoders (I2C) to stabilise and precisely point a high-inertia antenna tracker.

**Member, Ground Station Team | SS AMU SAT (university satellite project)** Sep 2024 – Present

- Built the tracker's 3D chassis model and two-axis PID positioning logic; support satellite tracking operations and calibration of communication equipment.

## PROJECTS

- UHF Satellite Image Transfer & SDR Testing** (2026): 435 MHz SDR link for image transfer with BPSK pipelines in GNU Radio and Python telemetry scripts.
- STM32 Test Board** (2026): Altium PCB with USB-C power delivery, buck converter and motion sensing, using hierarchical schematics.
- 437 MHz 9-Element Long-Yagi** (2026): DL6WU design fed directly at 50  $\Omega$  through a 1:1 balun, with a full construction manual.
- DC-DC Boost Converter** (2025): Simulink/Simscape analysis of steady-state and transient response across duty cycles and loads.
- Weather Dashboard** (2024): Node.js app on the OpenWeatherMap API, deployed on Render ([Live](#) | [Source](#)).

## SKILLS

- Control & Drives:** FOC, MTPA, flux weakening, DTC, SVPWM, adaptive PID, MRAC; **Power Electronics:** DC-DC converters, inverter PWM, gate drivers
- Embedded:** C/C++, STM32 (CubeIDE/CubeMX), TI C2000 (CCS), Arduino, 8051, I2C, UART; **RF:** Yagi/cross-Yagi design, MMANA-GAL, NEC, GNU Radio, SDR
- Software & Tools:** MATLAB, Simulink, Python (NumPy, Pandas, scikit-learn), OpenCV, JavaScript, Node.js, Git, Fusion 360 (CAD, FEA), Altium, KiCad

## CERTIFICATIONS & LEADERSHIP

- Certifications:** GE Aerospace Electrical Engineering Job Simulation, Forage (2026); Data Science Course, CodeWithHarry (2026); MATLAB Simulation and Control, IEEE SB AMU (2025); OpenCV Bootcamp (2024)
- Chairperson, IEEE Student Branch, AMU** (Aug 2026 – Present): lead the branch's office-bearer team, technical events and workshops.
- India Mobile Congress:** represented Orbitalink and showcased the company's field unit.